OR4023 (Optimization) Homework 1

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Question 1. Prove the open ball

$$B_R = \{y \in \mathbb{R}^n : \|y\| < R\}$$

is open and convex.

Answer. Let $y \in B_R$, so $||y|| < R$. Set $\epsilon = R - ||y|| > 0$. For any z with $||z - y|| < \epsilon$, we have

$$||z|| \leq ||z - y|| + ||y|| < \epsilon + ||y|| = R,$$

so $z \in B_R$. Thus, $B(y, \epsilon) \subseteq B_R$, and B_R is open.

For $y_1, y_2 \in B_R$, and $y = (1 - \lambda)y_1 + \lambda y_2$ with $\lambda \in [0, 1]$, we have

$$||y|| \leq (1 - \lambda)||y_1|| + \lambda||y_2|| < (1 - \lambda)R + \lambda R = R.$$

So, $y \in B_R$, and B_R is convex.

Question 2. Prove the intersection of two convex sets is convex.

Answer. Let A and B be convex sets in $\mathbb{R}^n$. To show $A \cap B$ is convex, take any $x, y \in A \cap B$, so $x, y \in A$ and $x, y \in B$. Consider $z = (1 - \lambda)x + \lambda y$ where $\lambda \in [0, 1]$. Since A is convex,

$$z = (1 - \lambda)x + \lambda y \in A,$$

and since B is convex,

$$z = (1 - \lambda)x + \lambda y \in B.$$

Thus, $z \in A \cap B$. Hence, $A \cap B$ is convex.

Question 3. Find the first and second order Taylor series expansion of the function

$$f(x, y) = xe^y + \sin(xy)$$

at $(0,0)$ and estimate the error for the first order approximation.

Answer. First-order derivatives:

$$f_x = e^y + y\cos(xy), \quad f_y = xe^y + x\cos(xy).$$

Evaluating at $(0,0)$:

$$f_x(0,0) = e^0 + 0 = 1, \quad f_y(0,0) = 0 + 0 = 0.$$

Second-order derivatives:

$$f_{xx} = -y^2\sin(xy), \quad f_{xy} = e^y + \cos(xy) - xy\sin(xy), \quad f_{yy} = xe^y - x^2\sin(xy).$$

Evaluating at $(0,0)$:

$$f_{xx}(0,0) = 0, \quad f_{xy}(0,0) = e^0 + 1 = 2, \quad f_{yy}(0,0) = 0.$$

First-order Taylor expansion:

$$f(x,y) \approx f(0,0) + f_x(0,0)x + f_y(0,0)y = x.$$

Second-order Taylor expansion:

$$f(x,y) \approx x + \frac{1}{2}(4xy) = x + 2xy.$$

The error term in the first-order Taylor approximation is given by

$$R_1(x,y) = \frac{1}{2}[f_{xx}(\xi,\eta)x^2 + 2f_{xy}(\xi,\eta)xy + f_{yy}(\xi,\eta)y^2],$$

for some (ξ,η) on the line segment joining $(0,0)$ and (x,y). We can estimate

$$|f_{xx}(\xi,\eta)| \leqslant \eta^2 \leqslant y^2, \quad |f_{xy}(\xi,\eta)| \leqslant e^\eta + 1 + \xi\eta \leqslant e^y + 1 + xy,$$

$$|f_{yy}(\xi,\eta)| \leqslant \xi e^\eta + \xi^2 \leqslant xe^y + x^2.$$

Thus

$$|R_1(x,y)| \leqslant \frac{1}{2}[y^2x^2 + 2(e^y + 1 + xy)xy + (xe^y + x^2)y^2].$$

Remark 1. If x, y is close to $(0,0)$, then the RHS is approximately $2xy$, which is the second term in the Taylor expansion.

Question 4. Prove that if $f(x)$ and $g(x)$ are two convex functions defined on a convex set $S \subseteq \mathbb{R}^n$, then $f(x) + g(x)$ is also a convex function on S.

Answer. A function $f(x)$ is convex on S if for all $x_1, x_2 \in S$ and $\lambda \in [0,1]$, we have

$$f((1-\lambda)x_1 + \lambda x_2) \leq (1-\lambda)f(x_1) + \lambda f(x_2).$$

Similarly, since $g(x)$ is convex, we have

$$g((1-\lambda)x_1 + \lambda x_2) \leq (1-\lambda)g(x_1) + \lambda g(x_2).$$

Let $h = f + g$, then

$$h((1-\lambda)x_1 + \lambda x_2) \leq (1-\lambda)h(x_1) + \lambda h(x_2),$$

which shows that $h(x)$ is convex.

Question 5. Consider the function $f\colon \mathbb{R} \to \mathbb{R}$ defined piecewise as follows:

$$f(x) = \begin{cases} 0 & \text{if } x \leq 0, \\ x^2 & \text{if } x > 0. \end{cases}$$

a) Show that f is convex by definition.

b) Show that f is differentiable everywhere.

c) Show that f is convex by the first derivative test.

Answer. Answer.

a) **Convexity by Definition:**

To prove convexity using the definition, we must show that for all $x_1, x_2 \in \mathbb{R}$ and $\lambda \in [0,1]$,

$$f((1-\lambda)x_1 + \lambda x_2) \leq (1-\lambda) f(x_1) + \lambda f(x_2).$$

- If $x_1, x_2 \leq 0$, then $f(x_1) = f(x_2) = 0$, and

$$f((1-\lambda)x_1 + \lambda x_2) = 0 = (1-\lambda) f(x_1) + \lambda f(x_2).$$

- If $x_1, x_2 > 0$, then $f(x) = x^2$, and we check

$$f((1-\lambda)x_1 + \lambda x_2) = ((1-\lambda)x_1 + \lambda x_2)^2.$$

  which holds since f is convex for $x \in \mathbb{R}$.

- If $x_1 \leq 0$ and $x_2 > 0$, then let $(1-\lambda) f(x_1) + \lambda f(x_2) = \lambda x_2^2$. Consider two subcases
  - if $(1-\lambda)x_1 + \lambda x_2 \leqslant 0$, then

$$f((1-\lambda)x_1 + \lambda x_2) = 0 \leqslant \lambda x_2^2.$$

  - if $(1-\lambda)x_1 + \lambda x_2 > 0$, then

$$f((1-\lambda)x_1 + \lambda x_2) = [(1-\lambda)x_1 + \lambda x_2]^2 \leqslant (\lambda x_2)^2 \leqslant \lambda x_2^2.$$

  In both subcases, we have

$$f((1-\lambda)x_1 + \lambda x_2) \leq (1-\lambda) f(x_1) + \lambda f(x_2).$$

Thus, $f(x)$ is convex by definition.

b) **Differentiability:**

We compute the derivative.

- For $x > 0$,

$$f'(x) = 2x.$$

- For $x < 0$,

$$f'(x) = 0.$$

- At $x = 0$, we check the left and right derivatives:

$$\lim_{h \to 0^-} \frac{f(h) - f(0)}{h} = \lim_{h \to 0^-} \frac{0 - 0}{h} = 0.$$

$$\lim_{h \to 0^+} \frac{f(h) - f(0)}{h} = \lim_{h \to 0^+} \frac{h^2 - 0}{h} = \lim_{h \to 0^+} h = 0.$$

Since both one-sided derivatives exist and are equal, $f'(0) = 0$.

c) **Convexity by First Derivative Test:**

We want to show

$$f(y) \geqslant f(x) + f'(x)(y - x)$$

for all x, y. First, assume $y > x$.

- If $x, y \leqslant 0$, then $f(x) = f(y) = f'(x) = 0$, and the inequality holds.
- If $x \leqslant 0, y > 0$, then $f(x) = 0$, $f(y) = y^2$, and $f'(x) = 0$. So the inequality holds.
- If $x > 0, y > 0$, then $f(x) = x^2$, $f(y) = y^2$, $f'(x) = 2x$. So

$$f(x) + f'(x)(y - x) = x^2 + 2x(y - x) = 2xy - x^2 \leqslant 2y^2 - y^2 = y^2 = f(y).$$

The proof for the case $x > y$ is omitted.

Question 6. Let $f(x_1, x_2) = x_1^2 - x_1 x_2 + x_2^2$. Is f convex on $\mathbb{R}^2$?

Answer. We have

$$\nabla f = \begin{pmatrix} 2x_1 - x_2 \\ -x_1 + 2x_2 \end{pmatrix}, \quad H = \nabla^2 f = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}.$$

The principal minors of H are $2, 3$. Hence H is positive definite, so f is convex.

Question 7. Fit the data points $(0, 1), (1, 3), (2, 2)$ by a linear function $y(t) = x_1 t + x_2$ using the least squares method with the following steps.

a) Formulate an optimization problem.

b) Find the stationary point and show it is a global minimizer.

c) Plot a graph to illustrate your result.

Answer.

a) Let $t_1 = 0, t_2 = 1, t_3 = 2$ and $y_1 = 1, y_2 = 3, y_3 = 2$. Define

$$f(x_1, x_2) = \sum_{i=1}^{3} (y_i - y(t_i))^2 = (1 - x_2)^2 + [3 - (x_1 + x_2)]^2 + [2 - (2x_1 + x_2)]^2.$$

Then the optimization problem is

$$\min_{(x_1, x_2) \in \mathbb{R}^2} f(x_1, x_2).$$

b) We have

$$\nabla f = \begin{pmatrix} -2[3 - (x_1 + x_2)] - 4[2 - (2x_1 + x_2)] \\ -2(1 - x_2) - 2[3 - (x_1 + x_2)] - 2[2 - (2x_1 + x_2)] \end{pmatrix} = \begin{pmatrix} 10x_1 + 6x_2 - 14 \\ 6x_1 + 6x_2 - 12 \end{pmatrix}.$$

Solving $\nabla f = 0$ yields $x_1 = \frac{1}{2}, x_2 = \frac{3}{2}$. Moreover,

$$\nabla^2 f = \begin{pmatrix} 10 & 6 \\ 6 & 6 \end{pmatrix}$$

is positive definite since its principal minors are 10 and $60 - 36 = 24$. Hence f is convex and the stationary point is a global minimizer.

c) use https://www.desmos.com/calculator

